

## LISMORE, Australia

WMO: 945720

Lat: 28.8306S

Lon: 153.2600E

Elev: 31

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |          | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                          |          |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD     |              |
| (a)<br>7      | (b)<br>38.0 | (c)<br>40.5 | (d)<br>28.2                   | (e)<br>22.1 | (f)<br>60.4 | (g)<br>31.7 | (h)<br>26.1 | (i)<br>58.5 | (j)<br>20.5           | (k)<br>64.0 | (l)<br>18.5 | (m)<br>63.8 | (n)<br>5.4               | (o)<br>0 | (p)<br>0.379 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>1      | (b)<br>18.3            | (c)<br>93.5     | (d)<br>74.2 | (e)<br>90.1 | (f)<br>73.1 | (g)<br>87.2 | (h)<br>72.4 | (i)<br>77.9         | (j)<br>88.4 | (k)<br>76.3 | (l)<br>85.6 | (m)<br>75.0 | (n)<br>83.4 | (o)<br>10.8          | (p)<br>350 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.0                            | 131.4 | 82.3 | 73.6 | 125.4 | 81.2 | 72.4 | 120.1 | 79.8 | 41.4          | 88.5 | 39.9 | 85.4 | 38.5 | 83.5 |                   |
|                                 |       |      |      |       |      |      |       |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 19.5              | 17.4 | 15.7 | 32.9                       | 101.1 | 2.0                | 3.6 | 31.4                                               | 103.7 | 30.3       | 105.8 | 29.1       | 107.8 | 27.7       | 110.4 |
|                   |      |      | 31.7                       | 79.9  | 2.5                | 1.9 | 29.9                                               | 81.3  | 28.4       | 82.4  | 27.0       | 83.5  | 25.2       | 84.9  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 66.6   | 75.5  | 74.8  | 72.5  | 67.6  | 61.6  | 57.8  | 56.4  | 58.2  | 63.3  | 67.3  | 70.9  | 73.5  |
|                                                                        | DBStd       | 7.82   | 3.97  | 3.86  | 3.31  | 3.16  | 3.90  | 4.05  | 3.93  | 4.10  | 4.50  | 4.72  | 4.60  | 4.43  |
|                                                                        | HDD50       | 3      | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 944    | 0     | 0     | 0     | 10    | 119   | 216   | 267   | 215   | 84    | 28    | 4     | 1     |
|                                                                        | CDD50       | 6055   | 789   | 695   | 697   | 528   | 359   | 236   | 201   | 256   | 400   | 537   | 628   | 729   |
|                                                                        | CDD65       | 1520   | 324   | 275   | 232   | 88    | 13    | 1     | 2     | 5     | 34    | 100   | 181   | 265   |
|                                                                        | CDH74       | 11838  | 2635  | 2041  | 1335  | 453   | 117   | 10    | 27    | 144   | 500   | 953   | 1526  | 2097  |
|                                                                        | CDH80       | 3729   | 942   | 676   | 324   | 57    | 10    | 1     | 1     | 23    | 133   | 307   | 518   | 737   |
| (14) Wind                                                              | WSAvg       | 6.7    | 7.2   | 6.8   | 5.9   | 5.6   | 5.5   | 6.1   | 6.4   | 7.1   | 7.4   | 7.7   | 7.8   | 7.5   |
| (15) Precipitation                                                     | PrecAvg     | 55.6   | 5.8   | 7.3   | 7.8   | 5.7   | 4.8   | 4.1   | 3.0   | 2.0   | 1.5   | 3.4   | 4.1   | 5.6   |
|                                                                        | PrecMax     | 82.4   | 18.1  | 19.5  | 26.4  | 20.8  | 18.1  | 24.0  | 18.1  | 7.6   | 8.4   | 20.9  | 9.6   | 18.2  |
|                                                                        | PrecMin     | 29.6   | 0.8   | 1.9   | 1.5   | 1.0   | 0.6   | 0.3   | 0.1   | 0.0   | 0.0   | 0.4   | 1.0   | 0.9   |
|                                                                        | PrecStd     | 14.7   | 4.0   | 4.7   | 5.6   | 5.0   | 4.1   | 4.7   | 3.9   | 1.6   | 1.7   | 3.9   | 2.4   | 4.1   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 97.5  | 96.2  | 91.8  | 85.0  | 80.8  | 75.2  | 77.5  | 83.1  | 90.0  | 92.4  | 95.4  | 97.1  |
|                                                                        |             | MCWB   | 77.3  | 78.0  | 74.8  | 71.0  | 66.8  | 63.0  | 61.3  | 61.2  | 64.2  | 68.7  | 70.7  | 73.8  |
|                                                                        | 2%          | DB     | 91.9  | 90.4  | 86.4  | 81.3  | 77.0  | 72.0  | 73.4  | 77.9  | 83.4  | 87.3  | 90.0  | 91.5  |
|                                                                        |             | MCWB   | 75.3  | 76.1  | 73.4  | 69.0  | 64.5  | 60.8  | 58.6  | 60.0  | 64.1  | 67.8  | 70.7  | 73.1  |
|                                                                        | 5%          | DB     | 88.2  | 87.0  | 83.3  | 78.8  | 74.4  | 69.5  | 70.1  | 74.1  | 79.4  | 83.2  | 85.6  | 87.1  |
|                                                                        |             | MCWB   | 74.5  | 74.3  | 72.0  | 68.0  | 63.2  | 59.6  | 58.3  | 58.6  | 63.0  | 66.9  | 70.7  | 72.5  |
|                                                                        | 10%         | DB     | 85.1  | 83.8  | 80.7  | 76.5  | 71.7  | 67.0  | 67.0  | 70.5  | 75.9  | 79.2  | 82.0  | 83.8  |
|                                                                        |             | MCWB   | 74.0  | 72.6  | 70.7  | 67.3  | 61.9  | 58.8  | 56.5  | 57.2  | 62.1  | 66.1  | 69.1  | 71.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 79.9  | 80.3  | 76.7  | 73.7  | 70.5  | 66.5  | 65.9  | 66.8  | 69.5  | 74.4  | 75.7  | 78.3  |
|                                                                        |             | MCDB   | 91.7  | 91.9  | 86.5  | 80.8  | 75.9  | 71.1  | 72.8  | 77.7  | 82.6  | 85.9  | 88.8  | 91.4  |
|                                                                        | 2%          | WB     | 78.0  | 77.9  | 75.1  | 71.4  | 67.8  | 64.1  | 62.7  | 63.0  | 67.3  | 71.3  | 73.6  | 76.0  |
|                                                                        |             | MCDB   | 87.8  | 87.9  | 83.2  | 77.7  | 73.5  | 68.2  | 68.0  | 72.2  | 78.9  | 81.1  | 85.1  | 86.9  |
|                                                                        | 5%          | WB     | 76.3  | 76.2  | 73.8  | 70.0  | 65.5  | 62.3  | 60.5  | 61.1  | 65.6  | 69.2  | 72.0  | 74.3  |
|                                                                        |             | MCDB   | 84.9  | 84.4  | 80.6  | 76.4  | 71.6  | 66.9  | 66.2  | 69.6  | 75.2  | 78.2  | 82.0  | 83.8  |
|                                                                        | 10%         | WB     | 74.9  | 74.3  | 72.4  | 68.4  | 63.8  | 60.7  | 58.7  | 59.3  | 63.9  | 67.4  | 70.6  | 72.8  |
|                                                                        |             | MCDB   | 82.8  | 81.3  | 78.6  | 74.3  | 69.1  | 65.0  | 64.4  | 67.2  | 72.6  | 76.1  | 79.5  | 81.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 18.3  | 17.7  | 17.2  | 18.0  | 22.0  | 19.9  | 23.3  | 26.4  | 25.6  | 22.6  | 21.1  | 19.5  |
|                                                                        |             | MCDBR  | 25.2  | 23.6  | 22.1  | 22.7  | 25.1  | 23.2  | 28.0  | 32.5  | 31.8  | 31.2  | 28.3  | 26.6  |
|                                                                        |             | MCWBR  | 11.0  | 10.5  | 10.1  | 11.6  | 13.5  | 13.1  | 15.1  | 16.0  | 14.5  | 13.7  | 12.0  | 11.2  |
|                                                                        | 5% WB       | MCDBR  | 21.2  | 21.3  | 18.6  | 19.7  | 20.6  | 17.1  | 20.6  | 25.8  | 26.2  | 24.3  | 23.2  | 22.0  |
|                                                                        |             | MCWBR  | 9.9   | 9.9   | 9.2   | 11.5  | 12.9  | 11.1  | 13.1  | 14.4  | 13.4  | 11.7  | 10.8  | 10.1  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.388 | 0.373 | 0.356 | 0.349 | 0.314 | 0.309 | 0.300 | 0.313 | 0.353 | 0.379 | 0.384 | 0.394 |
|                                                                        | taud        |        | 2.488 | 2.545 | 2.587 | 2.570 | 2.638 | 2.642 | 2.651 | 2.594 | 2.476 | 2.433 | 2.464 | 2.456 |
|                                                                        | Ebn at Noon |        | 302   | 302   | 297   | 284   | 281   | 275   | 283   | 292   | 293   | 296   | 302   | 301   |
|                                                                        | Edh at Noon |        | 37    | 34    | 31    | 29    | 24    | 23    | 23    | 27    | 34    | 37    | 37    | 38    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2049  | 1859  | 1578  | 1340  | 1109  | 919   | 1065  | 1349  | 1675  | 1871  | 2023  | 2038  |
|                                                                        | RadStd      |        | 169   | 153   | 89    | 65    | 69    | 59    | 84    | 103   | 133   | 169   | 172   | 206   |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (21 neighbors) | +0.70 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -104  | +304  | +187  |

Nomenclature: See separate page